

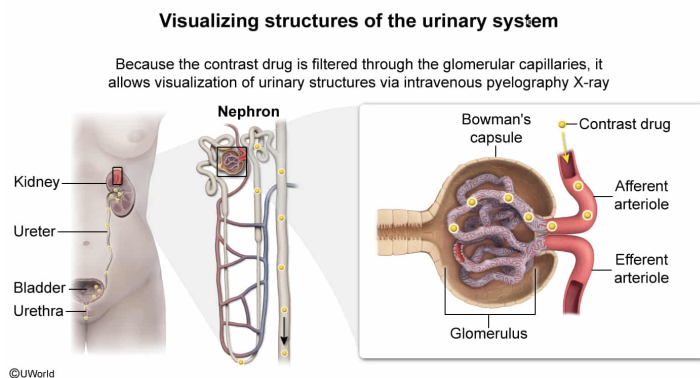
Intravenous pyelography (IVP) is a procedure in which a contrast drug containing iodine is injected into a patient and subsequent visualization of urinary structures becomes possible through X-ray imaging. Which characteristic best describes these contrast drugs? Each drug:

- ☐ A. is reabsorbed by the proximal convoluted tubule. [14%]
- ☐ B. travels in the bloodstream while bound to albumin. [10%]
- ✓ ☒ C. is filtered through the glomerular capillaries. [63%]
- ☐ D. drains into the efferent arterioles of nephrons. [11%]

Correct

63% answered correctly

Explanation:



Several structures make up the **urinary system**, including the **kidneys**, ureters, bladder, and urethra. The function of this system is to filter the **blood** and **produce liquid waste** (ie, urine) that is **excreted** from the body.

The kidney is divided into small functional units (**nephrons**) that are responsible for filtering blood and concentrating urine. Blood to be filtered initially enters the kidneys via the renal **arteries**, which branch into small afferent arterioles. These arterioles themselves branch into the **glomerulus**, or a ball-like network of capillaries.

The glomerular capillaries contain large pores through which all noncellular blood components pass as blood is filtered. The capillaries are lined by a basement membrane that allows the passage of water, most solutes, and very small proteins into the filtrate. These filtered

substances are collected by the Bowman's capsule, which surrounds the glomerulus, and are delivered to the subsequent nephron section (ie, proximal convoluted tubule). All materials not filtered by the glomerulus exit via the efferent arterioles.

In this question, intravenous pyelography (IVP) is a procedure in which a contrast drug containing iodine is injected into a patient. This allows subsequent visualization of urinary structures (ie, kidneys, ureters,

bladder, urethra) through X-ray imaging. For this visualization to occur, the drugs must travel in the urine through the urinary structures.

Therefore, each drug must be **filtered through the glomerular capillaries**, allowing it to become part of the filtrate that will eventually be excreted as urine.

The other characteristics would *not* allow visualization of urinary structures because they would cause the contrast drugs to be found in the bloodstream, *not* in the filtrate or urine:

- **(Choice A)** If the contrast drugs are reabsorbed by the proximal convoluted tubule, they will reenter the general circulation rather than remain in the filtrate.
- **(Choice B)** Albumin is a plasma protein that allows the capillaries to maintain their oncotic pressure. Because of its relatively large size, it is not filtered through the glomerulus and thus remains in the blood. Therefore, if the contrast drugs traveled in the bloodstream while bound to albumin, they would not become part of the filtrate.
- **(Choice D)** If the contrast drugs drained into the efferent arterioles of nephrons, this means they were not filtered through the glomerular capillaries; rather, they remained in the bloodstream.

Educational objective:

Substances filtered through the glomerular capillaries will be found in the urine unless they are reabsorbed by another portion of the nephron.